

AMENDMENT NO. 1 DECEMBER 1984
TO
IS : 8588 (Part I) - 1977 SPECIFICATION FOR
THERMOSTATIC BIMETALS

PART I GENERAL REQUIREMENTS AND
METHODS OF TESTS

Alterations

(Page 3, clause 0.3) — Substitute the following for the existing clause:

" 0.3 While preparing this standard, assistance has been derived from the following:

DIN 1715 ' Thermostatic bimetals ', Deutscher Normenausschuss

ASTM B106-78 ' Test for Flexivity of Thermostat Metals '
American Society for Testing Materials "

(Page 4, clause 2.2, para 2) — Substitute the following for the existing para:

' A thermostatic bimetal having a thickness t which is straight at a temperature T_1 , will assume at a temperature T_2 , a curved term governed by the following relationship:

$$K = \frac{1}{R} = \frac{2a}{t} (T_2 - T_1)$$

where

a is the specific thermal deflection (deg⁻¹), and

K is the curvature (mm⁻¹). '

[Page 4, clause 2.4 (renumber as 2.5)] — Substitute the following for the existing clause'

' **2.4 Linearity Range** — It denotes the temperature range in which the thermal deflection does not vary by more than ± 5 percent from the deflection, which is calculated from the flexivity (see Fig. 2) '

[Page 4, Note under clause 2.6 (renumber as 2.7)] — Substitute ' 27°C ' for ' 20°C '.

[Page 7, clause 9.1.1(a)] — Substitute ' Flexivity ' for ' Specific thermal deflection '.

(Page 7, clause 9.1.3, line 1) — Substitute ' Flexivity 'for ' Specific thermal deflection '.

(Page 7, clauses 9.2 and 9.2.1) — Substitute the following for the existing clauses:

" **9.2 Flexivity**

9.2.1 The test for the measurement of flexivity shall be conducted as given in IS : 2855-1964 ' Method of test for determining flexivity of thermostat metals '.

(Page 8, Fig. 1) — Substitute the following for the existing figure:

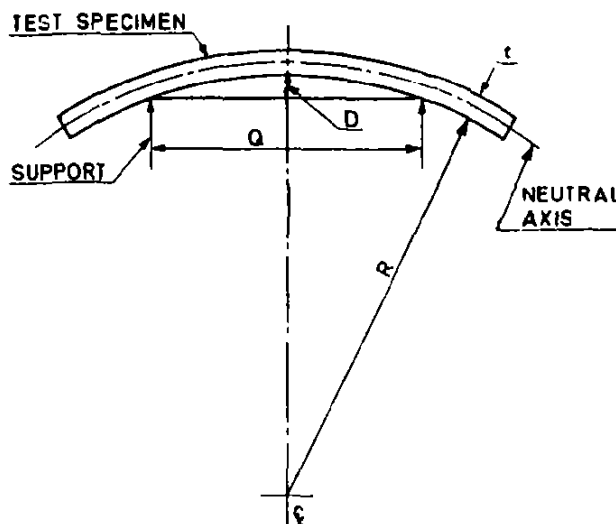


FIG 1 TEST FOR FLEXIVITY OF THERMOSTATIC BIMETALS

(Page 10, clause 9.4.1, para 2, lines 3 and 4) — Substitute ' thickness t ' for ' thickness s '.

(Page 10, clause 9.5) — Substitute the following for the existing clause:

9.5 Range of Linearity — This test shall be carried out according to 9.2. Measurement of deflection shall be done up to the maximum of specified linearity range at an interval of 20°C starting from 115°C. Two dotted straight lines as shown in Fig. 2 can be drawn by calculating deflection with the help of equation 4 of 2.4.'

Addenda

(Page 4, clause 2.3) — Add the following new clause after 2.3 and renumber the subsequent clauses accordingly:

" **2.4 Flexivity F** — It is the change of curvature of the longitudinal centre line of the specimen per unit temperature, change for unit thickness and is given by formula:

$$F = \frac{\left(\frac{1}{R_2} - \frac{1}{R_1} \right)}{T_2 - T_1} \times t \quad \dots (1)$$

where

F = flexivity,

R_2 = radius of curvature in mm at temperature T_2 ,

R_1 = radius of curvature in mm at temperature T_1 , and

t = thickness of test specimen in mm.

The curvature at either temperature is measured from the following formula:

$$\frac{1}{R} = \frac{8D}{Q^3 + D^3 + 4D^2} \quad \dots (2)$$

where

D = perpendicular distance in mm between centre line of the lower surface of the specimen and the straight line joining the support points (see Fig 1 or IS : 2855-1964 ' Method of test for determining flexivity of thermostat metals'), and

Q = distance between support points in mm.

The flexivity F can be derived from the equations (1) and (2).

$$F = \frac{8t}{T_2 - T_1} \left[\frac{D_2}{(Q^3 + 4D_2^2 + 4D_2^3)} - \frac{D_1}{(Q^3 + 4D_1^2 + 4D_1^3)} \right] \quad (3)$$

Considering the test specimen as flat and measurement carried out at a room temperature, flexibility F can be calculated from the following formula:

$$F = \frac{8Dt}{(T_2 - T_1)(Q^3 + 4Dt + 4D^2)} \quad \dots (4)$$

(ETDC 57)